TRANSFORMATIONS FOR DATA ANALYTICS(CPSC 4810)

Final Project

Due Date: July 30, 2025

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❒ This final group project should showcase your ability to collect, clean, transform, analyze, and visualize data to generate insights

❒ Your project should include the following components:

➦ Problem Statement Definition

➦ Data Collection

➦ Data Cleaning & Transformation

➦ Exploratory Data Analysis (EDA)

➦ Data Transformation & Feature Engineering

➦ Machine Learning Model (Prediction/Clustering)

➦ Dashboard

➦ Key Insights & Business Recommendation

➦ Git Version Control System (VCS) Usage

➦ Final Report & Presentation

Part-I
Problem Statement Definition

(5 marks)

- ❒ What is the specific business problem you are trying to solve?
- ❒ What are the scope, causes, and implications of the problem?
  - ➦ `Example: The company wants to understand customer behavior and predict future sales trends.`

Part-II
Data Collection

(5 marks)

- You may use data from various sources, including public datasets such as:
  - `Kaggle`
    - → https://www.kaggle.com/datasets
  - `University of California, San Diego (USD)` datasets
    - → https://datanexus.ucsd.edu/research-data/index.html
  - `University of California, Irvine (UCI)` datasets
    - → https://archive.ics.uci.edu/datasets
  - `company data`
  - `Web Application Interfaces (API)s`
  - ....

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Part-III
Data Cleaning & Transformation

(15 marks)

- Data cleaning involves several key steps:
  - `Addressing missing values by using techniques such as:`
    - dropna()
    - fillna ()
  - `Converting data types, including changing dates and categorical data to numeric formats`
  - `Eliminating duplicate entries to ensure data integrity`
  - `Normalizing or standardizing data to prepare it for modeling`
  - `ombining datasets through JOIN in SQL or using the merge() function in Python Pandas`

Part-IV
Exploratory Data Analysis (EDA)

(15 marks)

- Exploratory Data Analysis (EDA) should encompass the following components:
  - `Descriptive Statistics`
    - Mean, median, and standard deviation
  - `Visualizations`
    - `Histograms and boxplots to detect outliers`
    - `Correlation heatmap using seaborn.heatmap`
    - `Time-series analysis, such as monthly sales trends`
    - ....

Part-V
Data Transformation & Feature Engineering

(5 marks)

- Data Transformation and Feature Engineering should encompass the following steps:
  - `Create New Features`
    - `Example: Average Order Value: Total Revenue / Total Customers`
  - `Categorization`
    - `Example: Segment customers based on their spending habits`
  - `Scaling and Encoding`
  - `Example: Normalize numerical values and apply one-hot encoding to categorical values`

Part-VI
Machine Learning Model (Prediction/Clustering)

(5 marks)

- You have the option to use one of two machine learning models.
  - `The linear regression prediction model`
    - → `useful for understanding and predicting relationships between variables.`
  - `The K-Means clustering model`
  - → `useful for grouping similar data points into clusters based on their characteristics.`

Part-VII
Dashboard

(10 marks)

- ❒ The dashboard should feature an interactive visualization to enhance the presentation of your project data.
- ❒ You can choose from the following tools:
  - ➦ `Python`
    - → `using Matplotlib, Seaborn, or Plotly modules`
  - ➦ `R`
    - → `using gnuplot or Plotly packages`
  - ➦ `Tableau`
  - ➦ `Power BI`

Part-VIII
Key Insights & Business Recommendation

(10 marks)

- ❒ What are the most significant findings revealed by your data analysis project?
- ❒ What data-driven actions should be implemented?

Part-IX
Git Version Control System (VCS) Usage

(10 marks)

- ❒ Use Git VCS with a remote repos[1]

[1] Either you can use my remote server repos or use a different remote repos and provide me with access to it.

Part-X
Final Report & Presentation

(20 marks)

- ❒ Deliverables
  - ➦ Jupyter Notebook
  - ➦ Python/R scripts
  - ➦ Dashboard
- ❒ Presentation
  - ➦ Problem Statement
  - ➦ Data Overview
  - ➦ Methodology (EDA, ML Models)
  - ➦ Key Findings
  - ➦ Business Recommendations

Marking Scheme

Task	Marks
I- Problem Statement Definition	5
II- Data Collection	5
III- Data Cleaning & Transformation	15
IV- Exploratory Data Analysis (EDA)	15
V- Data Transformation & Feature Engineering	5
VI- Machine Learning Model (Prediction/Clustering)	5
VII- Dashboard	10
VIII- Key Insights & Business Recommendation	10
IX- Git Version Control System (VCS) Usage	10
X- Final Report & Presentation	20
Total	**100**

Submission

✎ Upload a zip file to Brightspace that contains your project deliverables and presentation.